Marketing Campaign Performance Analysis with SQL

Campaign reach, ROI, engagement, click-through rate, cost efficiency and channel performance across 200,005 campaigns

Tool: Microsoft SQL Server (SSMS) | 8 queries

1. Introduction

Companies spend heavily on online advertising across channels such as Google Ads, email, social media and video. Not every campaign pays off: some earn strong engagement and returns, while others waste budget. Reading campaign data well helps marketers decide where to spend, who to target and which channels to invest in.

This report uses SQL to turn a marketing campaign dataset into answers to practical questions. The analysis aims to:

- Check the reach of each campaign through impressions.
- Find the campaign with the highest ROI.
- Find the locations with the most impressions.
- Compare engagement across target audiences.
- Measure the overall click-through rate (CTR) and find campaigns with a strong CTR.
- Find the most cost-effective campaign.
- Rank marketing channels by estimated conversions.

2. Data Overview

The analysis uses one table, **campaigndata**, with **200,005 rows**. Each row is a single marketing campaign and Campaign_ID is the primary key. The data has no missing values and covers 5 companies, 6 channels, 5 target audiences and 5 locations. Dates were stored inconsistently in the original Excel file, so they were standardised before loading into SQL Server.

Column	Description
Campaign_ID	Unique identifier for each campaign (primary key)
Company	Company running the campaign
Campaign_Type	Type of campaign, such as Email, Display or Influencer
Target_Audience	Demographic group the campaign was aimed at
Duration	Length of the campaign, for example 30 days
Channel_Used	Email, Website, Google Ads, YouTube, Instagram or Facebook
Date	Date of the campaign
Customer_Segment	Customer interest group, such as Health & Wellness
Impressions	Times the ad was displayed (1,000 to 10,000)
Clicks	Times users clicked the ad (100 to 1,000)
Engagement_Score	Score from 1 to 10 for user interaction
Conversion_Rate	Share of clicks that led to the desired action, stored as a fraction (0.01 to 0.15)
Acquisition_Cost	Cost of acquiring customers through the campaign (5,000 to 20,000)
ROI	Return on investment (2.0 to 8.0)
Location	Region the campaign was aimed at

3. SQL Queries and Analysis

Query 1: Total impressions for each campaign

Method

SUM and GROUP BY total the impressions for each campaign, sorted from highest to lowest.

```
SELECT campaign_id,
       SUM(impressions) AS totalimpressions
FROM campaigndata
GROUP BY campaign_id
ORDER BY totalimpressions DESC;
```

Result

The campaigns at the top of the list each have 10,000 impressions, the highest value in the data. Across all 200,005 campaigns, impressions range from 1,000 to 10,000, with an average of about 5,500.

Insight

Because Campaign_ID is unique, this query lists each campaign's own impressions. Impressions only show reach, so they cannot tell strong campaigns from weak ones. Clicks, conversions and ROI are needed for that.

Query 2: Campaign with the highest ROI

Method

ORDER BY ... DESC with TOP 1 finds the campaign with the best return on investment.

```
SELECT TOP 1
       Campaign_ID,
       Company,
       ROI
FROM campaigndata
ORDER BY ROI DESC;
```

Result

The query returns a single campaign with an ROI of 8.0, for example Campaign 168 from NexGen Systems.

Insight

The highest ROI in the data is 8.0, but **160 campaigns share it**, so the campaign returned is just one of many ties and not a clear winner. TOP 1 hides ties, so a better question is what the 160 top campaigns have in common. Among them, NexGen Systems has the most (38), followed by Innovate Industries (33) and TechCorp (31).

Query 3: Top 3 locations by impressions

Method

SUM, GROUP BY and TOP 3 total the impressions for each location and keep the top three.

```
SELECT TOP 3
       Location,
       SUM(Impressions) AS TotalImpressions
FROM campaigndata
GROUP BY Location
ORDER BY TotalImpressions DESC;
```

Result

New York, Miami and Chicago have the most impressions.

Rank	Location	Total impressions
1	New York	221,359,756
2	Miami	221,347,726
3	Chicago	219,999,352

Insight

The gap between locations is tiny. Each of the five locations ran about 40,000 campaigns, and New York leads Los Angeles, the fourth-ranked location, by under 1%. Total impressions mostly reflect how many campaigns ran in each place, so no location clearly stands out.

Query 4: Average engagement score by target audience

Method

AVG and GROUP BY compare engagement across audiences.

```
SELECT target_audience,
       AVG(engagement_score) AS avgengagementscore
FROM campaigndata
GROUP BY target_audience
ORDER BY avgengagementscore DESC;
```

Result

Men 18-24 rank first, followed closely by the other four audiences.

Target audience	Average engagement score
Men 18-24	5.52
Women 25-34	5.49
Men 25-34	5.49
All Ages	5.49
Women 35-44	5.49

Insight

The gap between first place and the rest is only about 0.03 points on a 10-point scale. All five audiences engage at almost the same level, so audience choice alone does not explain engagement in this data.

Query 5: Overall click-through rate (CTR)

Method

The sum of clicks is divided by the sum of impressions and multiplied by 100.0 so SQL works in decimals and not whole numbers.

```
SELECT (SUM(Clicks) * 100.0 / SUM(Impressions)) AS OverallCTR
FROM campaigndata;
```

Result

The overall CTR is about **9.98%**.

Insight

Roughly 1 in 10 impressions leads to a click, which suggests the campaigns are fairly well targeted. There is still room to improve through better ad copy, visuals and calls to action.

Query 6: Most cost-effective campaign

Method

Cost per conversion is the acquisition cost divided by the number of conversions. Conversions are estimated as clicks multiplied by the conversion rate. Campaigns with no conversions are excluded to avoid dividing by zero, and results are sorted from lowest to highest cost.

```
SELECT TOP 1
       campaign_id,
       company,
       ROUND(CAST(acquisition_cost / (clicks * conversion_rate)
             AS DECIMAL(18,4)), 2) AS cost_per_conversion
FROM campaigndata
WHERE clicks * conversion_rate > 0
ORDER BY cost_per_conversion ASC;
```

Result

The most cost-effective campaign is **Alpha Innovations (Campaign ID 118451)**, at about **34.22** per conversion. It combines a low acquisition cost (5,107), a high number of clicks (995) and the top conversion rate (0.15).

Insight

Campaigns with a low cost per conversion get more customers for the same budget. This campaign combines a low cost with strong clicks and conversion rate, and no other campaign shares its exact value.

Query 7: Campaigns with a CTR above 5%

Method

A calculated CTR for each campaign (clicks * 100.0 / impressions) with a WHERE filter keeps only those above 5%.

```
SELECT campaign_id,
       company,
       (clicks * 100.0 / impressions) AS ctr
FROM campaigndata
WHERE (clicks * 100.0 / impressions) > 5;
```

Result

160,332 of the 200,005 campaigns (about 80%) have a CTR above 5%. For example, Innovate Industries (Campaign 1) has a CTR of 26.33%.

Insight

Most campaigns clear the 5% mark, so it is a low bar for this dataset. The highest CTRs are close to 99%, which is unrealistic for real ads and points to a simulated dataset. A higher cut-off would be more useful for picking out top performers.

Query 8: Channels ranked by estimated conversions

Method

Conversions for each campaign are estimated as clicks multiplied by the conversion rate, then added up for each channel. The RANK() window function orders the channels from most to fewest conversions.

```
SELECT channel_used,
       ROUND(CAST(SUM(clicks * conversion_rate)
             AS DECIMAL(18,2)), 0) AS total_conversions,
       RANK() OVER (ORDER BY SUM(clicks * conversion_rate) DESC)
             AS channel_rank
FROM campaigndata
GROUP BY channel_used
ORDER BY channel_rank;
```

Result

Email ranks first and Facebook last.

Rank	Channel	Estimated conversions
1	Email	1,485,394
2	Website	1,477,746
3	Google Ads	1,468,814
4	YouTube	1,463,621
5	Instagram	1,462,864
6	Facebook	1,446,295

Insight

The ranking mostly follows the number of campaigns each channel ran (Email 33,599, Facebook 32,820). Per campaign, every channel averages about 44 conversions, and Website is marginally the highest. So the channels perform almost identically, and a total-conversions ranking alone can mislead.

4. Key Findings

- The dataset has **200,005 campaigns**, with impressions from 1,000 to 10,000 each.
- The overall CTR is **9.98%**, and about 80% of campaigns have a CTR above 5%.
- The top ROI is 8.0, but **160 campaigns tie** for it, so no single campaign is uniquely the best.
- **Alpha Innovations (Campaign 118451)** is the most cost-effective campaign, at about 34.22 per conversion.
- **Email** leads on total estimated conversions and **Facebook** is last, but this follows campaign volume. Average conversions per campaign are about 44 for every channel.
- Locations and audiences perform very similarly. **Men 18-24** have the highest engagement (5.52), only 0.03 points ahead of the others.

5. Recommendations

1. **Study the 160 campaigns that tie for the top ROI.** Look at what they share on company, channel, audience and duration before copying any single campaign.
2. **Compare channels per campaign, not in total.** Total conversions mostly reflect how many campaigns a channel ran, so use average conversions or cost per conversion when moving budget.
3. **Copy the approach of Campaign 118451.** Its low acquisition cost, high clicks and top conversion rate make it the strongest cost-efficiency example.
4. **Raise the CTR bar.** A 5% threshold includes 80% of campaigns, so use a stricter cut-off such as the top 10% to find real leaders.
5. **Test audiences before shifting budget.** Engagement differs by only 0.03 points across audiences, so run A/B tests before favouring Men 18-24.
6. **Check the data before acting.** Very high CTRs and evenly spread group sizes suggest simulated data, so confirm the source before large budget decisions.

6. Limitations

- The group sizes and value ranges are very even, which suggests the dataset is synthetic, so results may not reflect real campaign behaviour.
- Differences between locations, audiences and channels are small, so rankings are weak evidence of real performance gaps.
- Conversions are **estimated** as clicks multiplied by the conversion rate, because the dataset has no conversion counts.
- Cost per conversion assumes Acquisition_Cost is the total cost of each campaign.
- Queries 2 and 6 use TOP 1, so ties for first place do not show. This matters for query 2, where 160 campaigns tie.

7. Skills Demonstrated

SELECT, GROUP BY, ORDER BY, WHERE, TOP, SUM / AVG, CAST, ROUND, calculated metrics (CTR, cost per conversion), RANK() OVER